

Skill Arena - Phase 8 Document 6

Mobile Application Specification, Notifications, Offline Replay System & Cross-Platform Experience Architecture

Document Purpose

Define the mobile experience, notification systems and cross-platform behaviour of Skill Arena.

Business Objective

Deliver a world-class mobile experience that matches the quality and functionality of the web platform.

Core Philosophy

Play anywhere, compete anywhere, review anywhere.

Supported Platforms

iOS, Android, Web and future tablet devices.

Mobile First Design

Interfaces optimized for touch interaction before desktop adaptation.

Authentication Experience

Secure login, biometric support and multi-factor authentication integration.

Mobile Dashboard

Quick access to rankings, wallet, challenges, tournaments and profile information.

Push Notification Framework

Challenge invites, tournament reminders, rewards, security alerts and social notifications.

Real-Time Notifications

Instant updates for matchmaking, challenge completion and tournament events.

Gameplay Experience

Responsive controls, low latency communication and server-authoritative validation.

Cross-Platform Synchronization

Player progress, rankings, wallets and replays synchronized across all devices.

Offline Replay System

Players may download approved replay packages for offline viewing.

Replay Download Controls

Replay downloads subject to security and platform policy controls.

Replay Sharing

Players may share replay links and approved replay highlights.

Replay Dispute Support

Replays can be attached to support tickets and integrity investigations.

Replay Analysis Tools

View routes, completion statistics and challenge history.

Challenge History

Players can review completed matches, house challenges and tournament activity.

Performance Optimization

Adaptive quality settings based on device capability.

Bandwidth Optimization

Efficient data transfer for mobile and low-bandwidth environments.

Accessibility Features

Scalable text, adaptive controls and future accessibility enhancements.

Security Requirements

Secure storage, device verification and anti-tampering protections.

Device Trust Integration

Mobile devices participate in trust score and fraud prevention systems.

Analytics Integration

Track engagement, retention, replay usage and device performance.

Monitoring Requirements

Monitor crashes, performance metrics and mobile user experience.

Future Expansion

Foundation for tablets, wearables, streaming integrations and future device categories.

Acceptance Criteria

Players receive a consistent, secure and high-quality experience across all supported platforms.